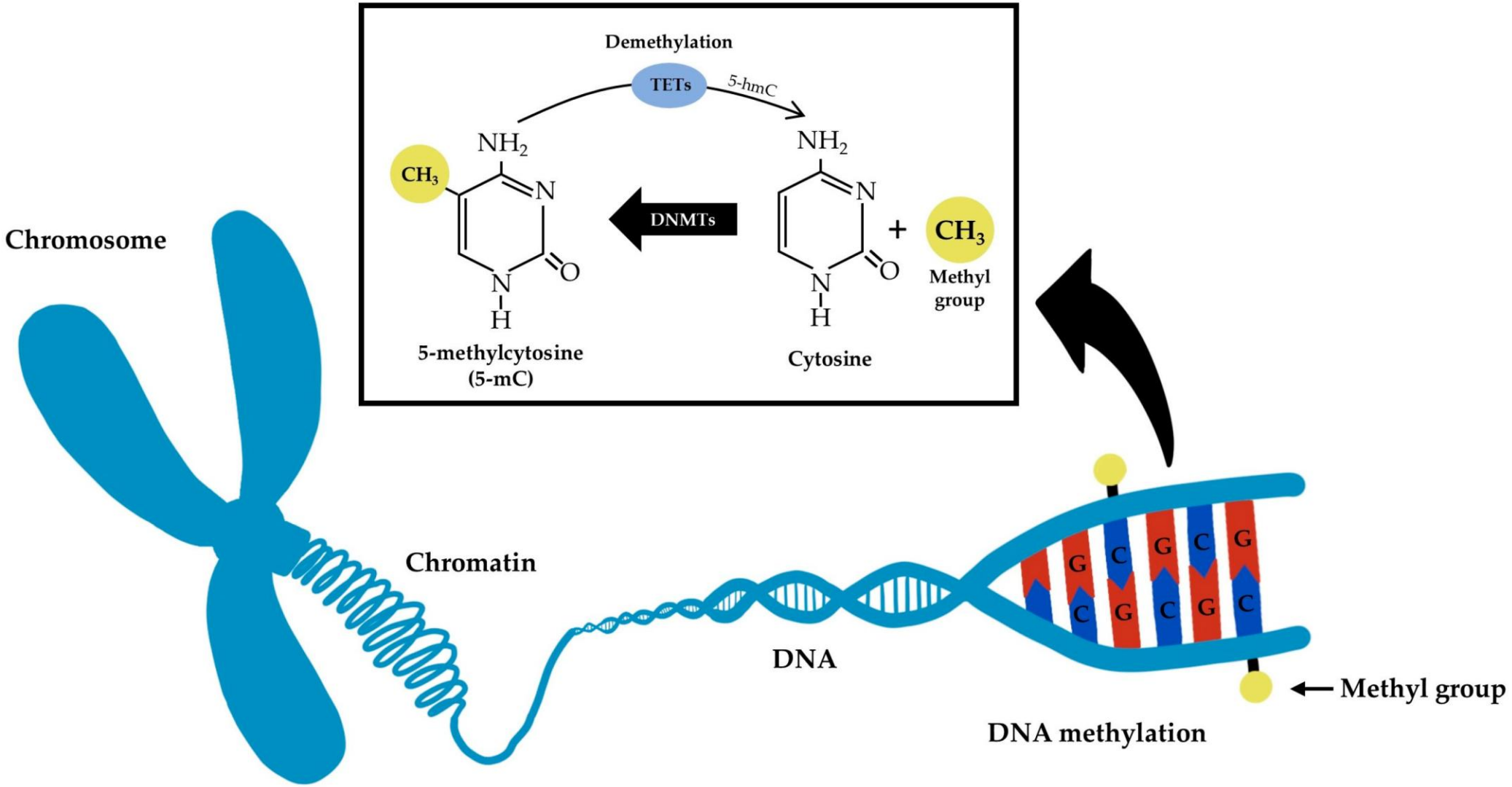
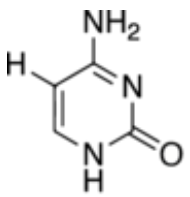
# L4

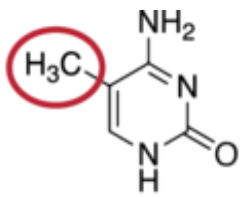
Adv Genomics

# Methylation of DNA






Cytosine




methylated Cytosine

- Methyl groups can be added to the Nts, changing their chemical and physical properties
- Cytosine and (very rarely) adenine can be methylated
- Methylation changes the activity of DNA without changing the sequence
- DNA methylation can be found in three sequence contexts:
  - CG (or CpG), most frequent
  - CHG or CHH (where H correspond to A, T or C)
- In mammals, around 3/4 of CpG dinucleotides are methylated by default
- Methylation occurs via a class of proteins called methyltransferases
- Methylation come at a cost, since over time deamination of 5-methylcytosine converts it to thymine, resulting in a T:G mismatch

- “CpG islands” are special places in the genome
- About 25,000 in the human genome
- CpG dinucleotides are underrepresented due to deamination
- In humans, C and G makes about 42% of the genome; you would expect a CpG frequency of 0.21*0.21= 0.441, but it is one-fifth of that
- Around 60-70% of human genes have a CpG island in their promoter region

CpG sites

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CCCGGGTCCGGGGCGGGGAAGAGCCGCCTCAACGGCAGGGCCCATCCGCGA
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GpC sites

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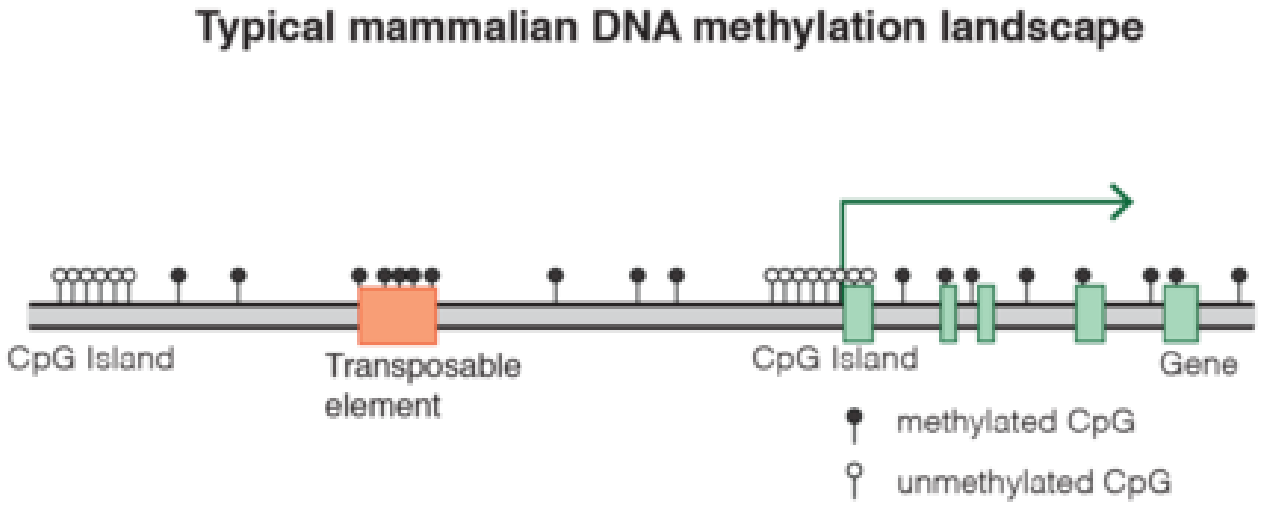
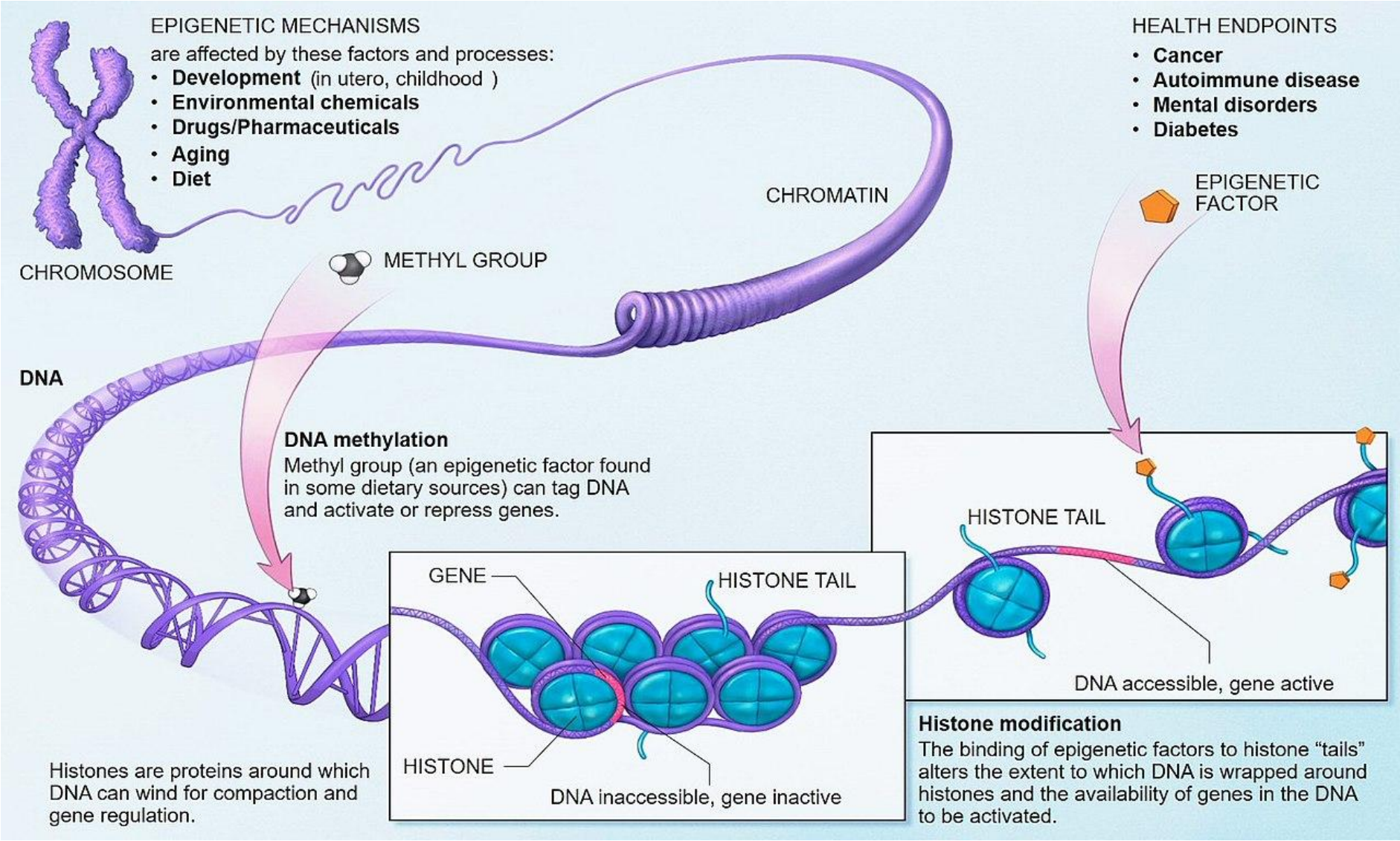
human *APRT* gene sequence

# DNA methylation affects gene transcription in two ways:

- The methylation of DNA itself may physically impede the binding of transcriptional proteins
- methylated DNA may be bound by methyl-CpG-binding domain proteins (MBDs) that recruit histone deacetylases and other chromatin remodeling proteins that form heterochromatin



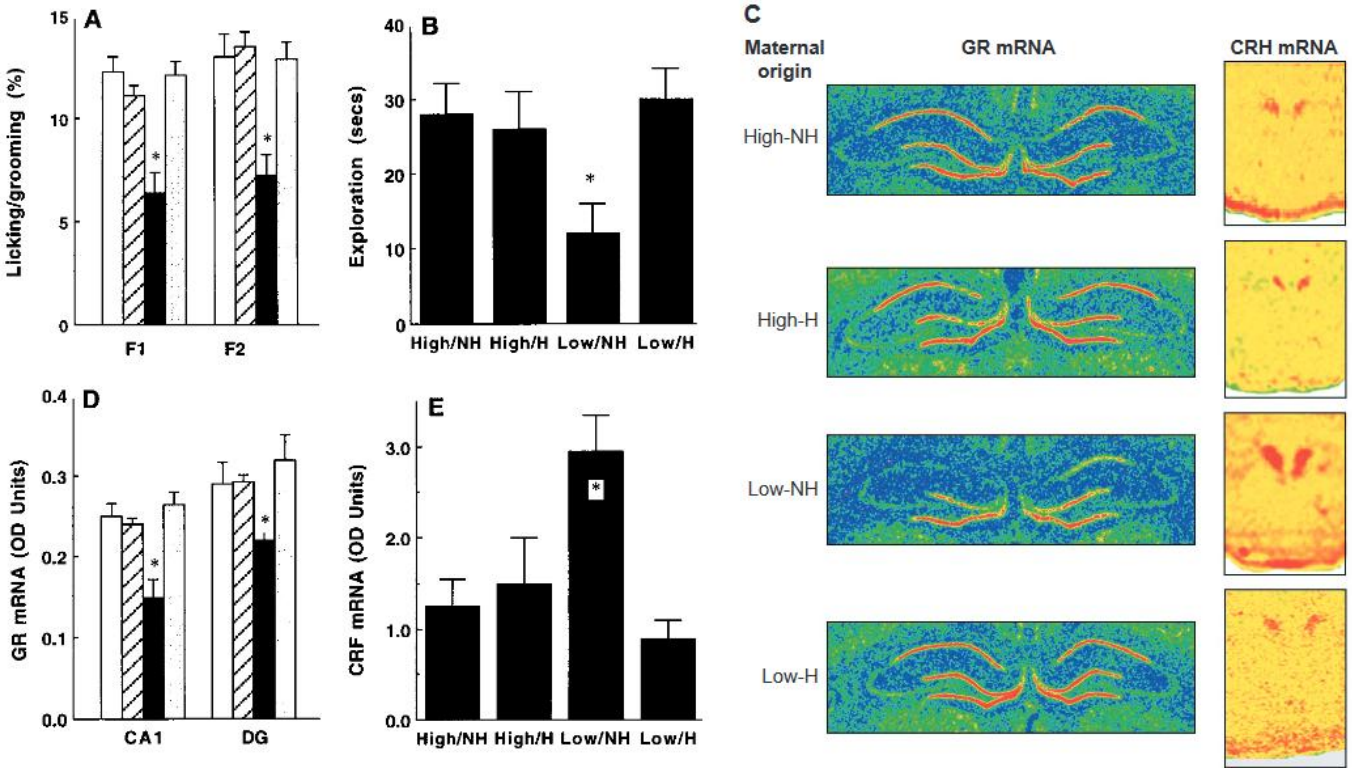


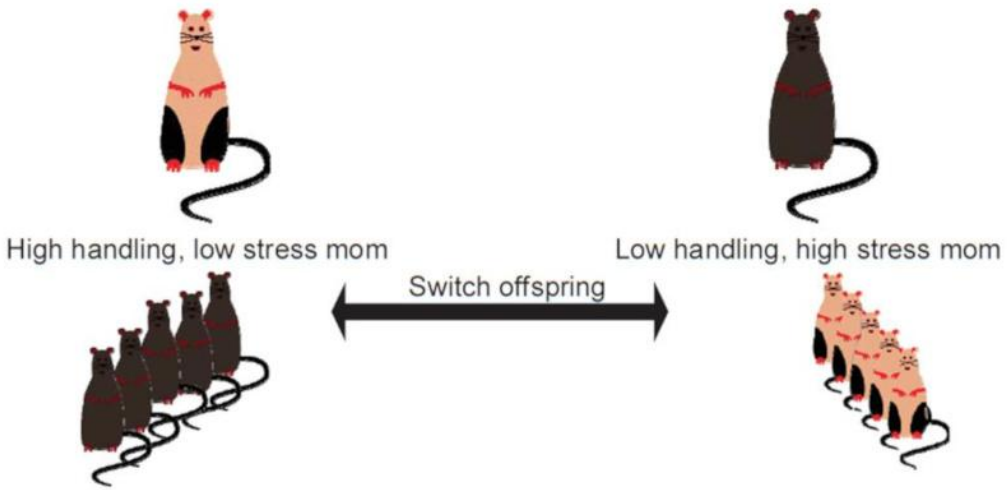

# Euchromatin VS Heterochromatin

# Nongenomic Transmission Across Generations of Maternal Behavior and Stress Responses in the Rat

Darlene Francis, Josie Diorio, Dong Liu, Michael J. Meaney*

In the rat, variations in maternal care appear to influence the development of behavioral and endocrine responses to stress in the offspring. The results of cross-fostering studies reported here provide evidence for (i) a causal relationship between maternal behavior and stress reactivity in the offspring and (ii) the transmission of such individual differences in maternal behavior from one generation of females to the next. Moreover, an environmental manipulation imposed during early development that alters maternal behavior can then affect the pattern of transmission in subsequent generations. Taken together, these findings indicate that variations in maternal care can serve as the basis for a nongenomic behavioral transmission of individual differences in stress reactivity across generations.



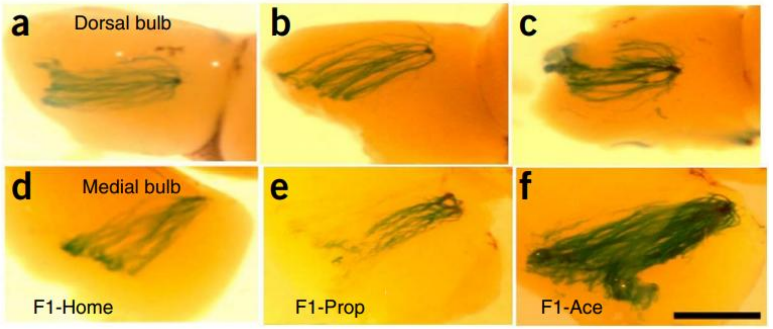



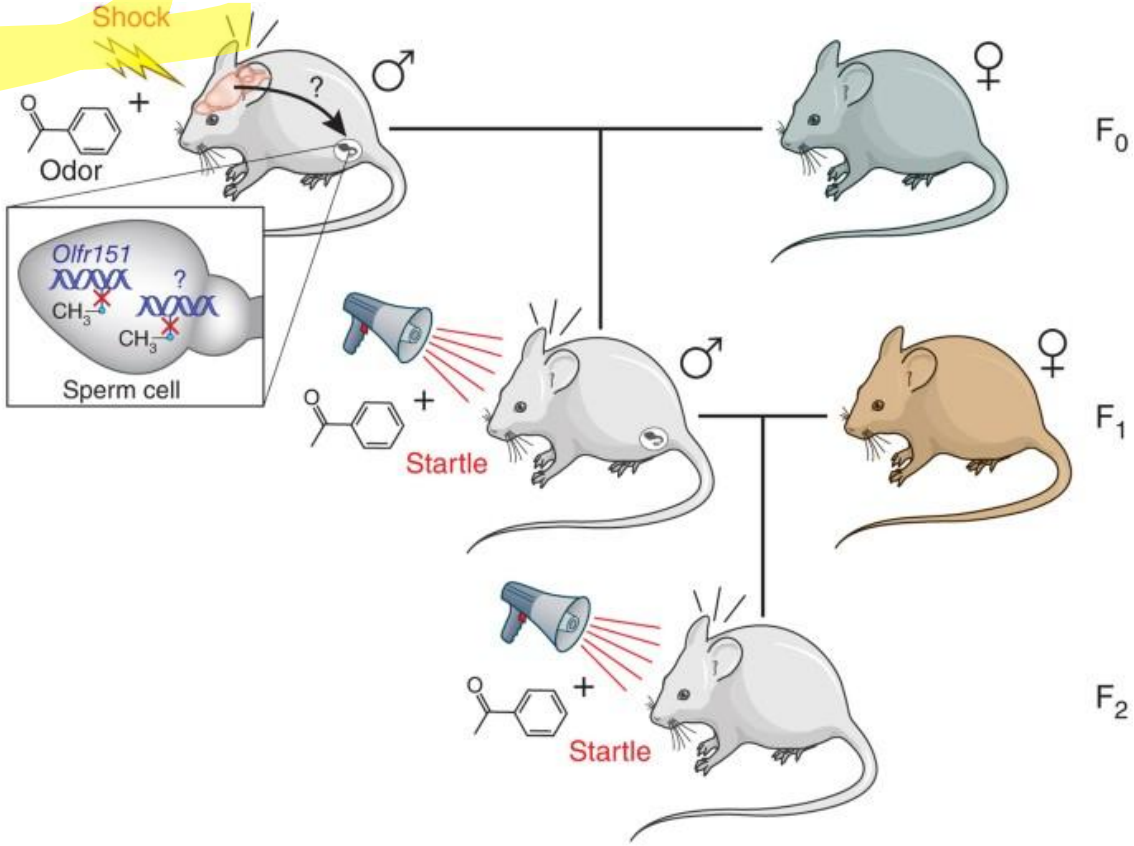

- Maternal care (licking) influences the activity of gene *NRC31* that protects against stress

# Parental olfactory experience influences behavior and neural structure in subsequent generations

Brian G Dias[1,2] & Kerry J Ressler[1–3]

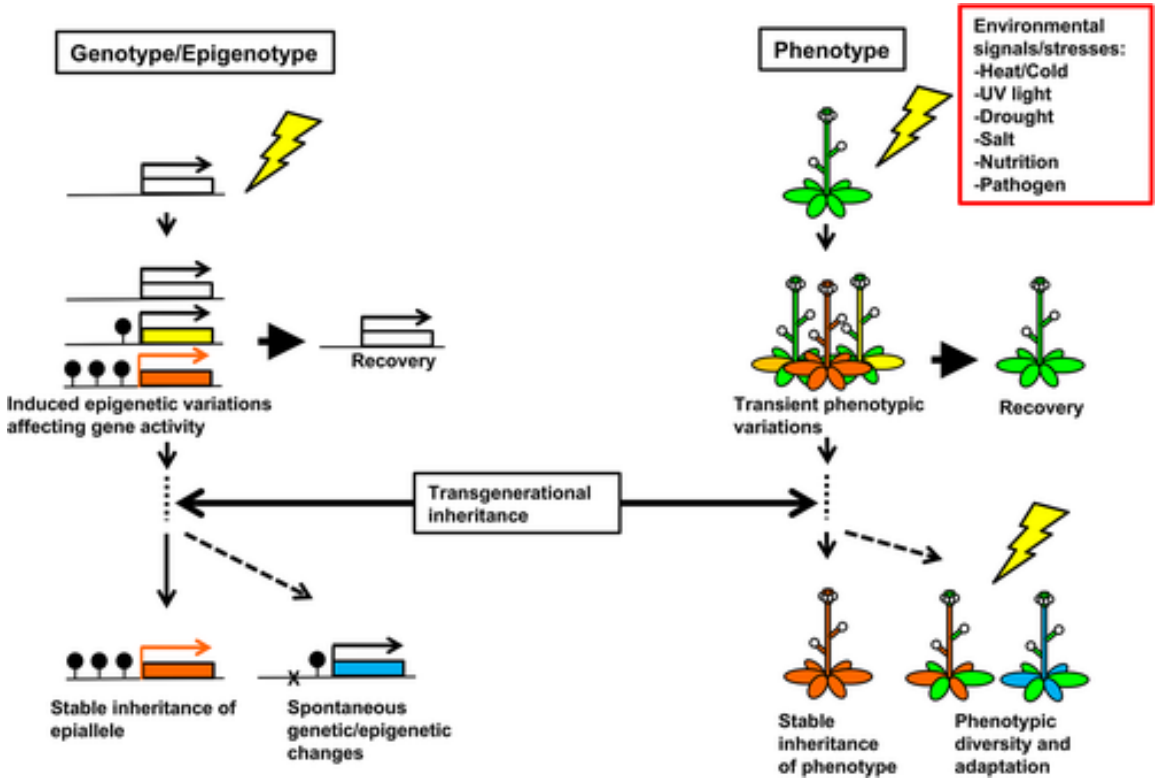
Using olfactory molecular specificity, we examined the inheritance of parental traumatic exposure, a phenomenon that has been frequently observed, but not understood. We subjected F0 mice to odor fear conditioning before conception and found that subsequently conceived F1 and F2 generations had an increased behavioral sensitivity to the F0-conditioned odor, but not to other odors. When an odor (acetophenone) that activates a known odorant receptor (*Olfr151*) was used to condition F0 mice, the behavioral sensitivity of the F1 and F2 generations to acetophenone was complemented by an enhanced neuroanatomical representation of the *Olfr151* pathway. Bisulfite sequencing of sperm DNA from conditioned F0 males and F1 naive offspring revealed CpG hypomethylation in the *Olfr151* gene. In addition, *in vitro* fertilization, F2 inheritance and cross-fostering revealed that these transgenerational effects are inherited via parental gametes. Our findings provide a framework for addressing how environmental information may be inherited transgenerationally at behavioral, neuroanatomical and epigenetic levels.




- Mice are imprinted to fear the odor acetophenone, which is recognized by *Olfr151*
- *Olfr151* is hypomethylated as a result
- Future generations have an enhanced response to acetophenone than controls

- Plantae are the most highly methylated eukaryotes, with up to 50% of their cytosine residues exhibiting methylation
- Being sessile organisms, gene expression regulation is pivotal to survival
- Plants possess a mechanism for active DNA demethylation involving DNA glycosylases that excise 5-meC and initiate its replacement with unmodified C through a **base excision repair (BER) pathway**

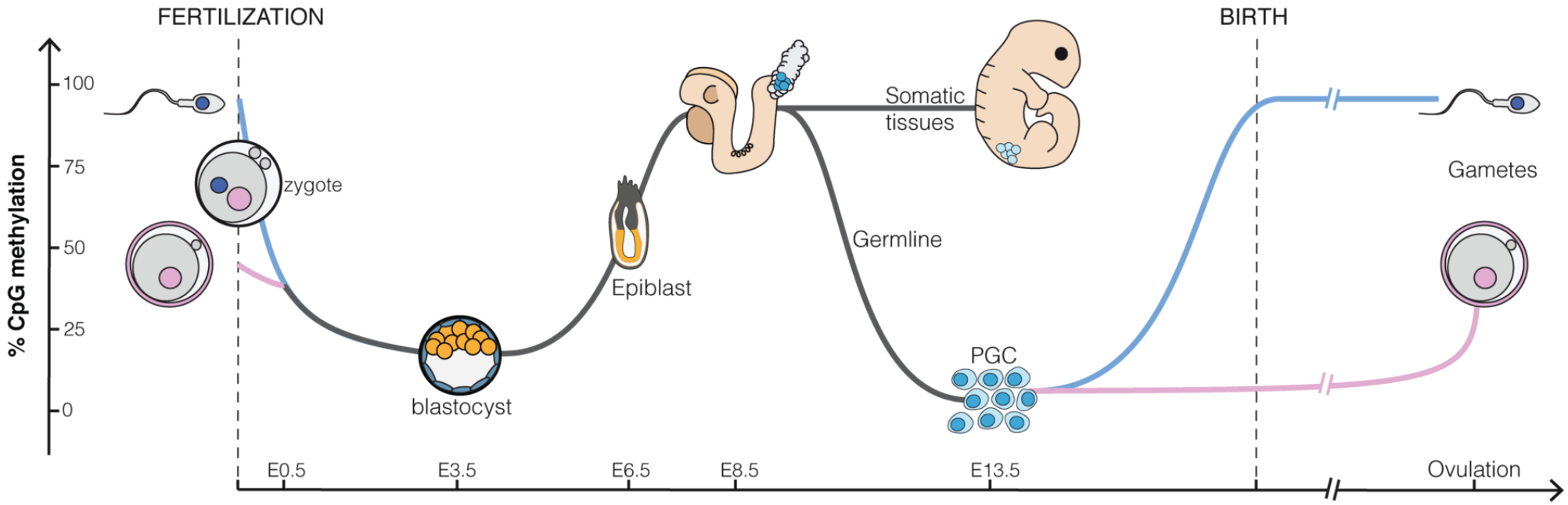



- Inherited epigenetic changes may become adaptive if offspring experiences environmental signals similarly to the previous generations
- If epigenetic changes are associated with genetic changes (e.g., transposon insertions) they can be stabilized as epialleles.
- Otherwise, epigenetic changes spontaneously revert to their initial states
- See https://www.nature.com/articles/cr201123.pdf

https://doi.org/10.1002/1438-390X.12018

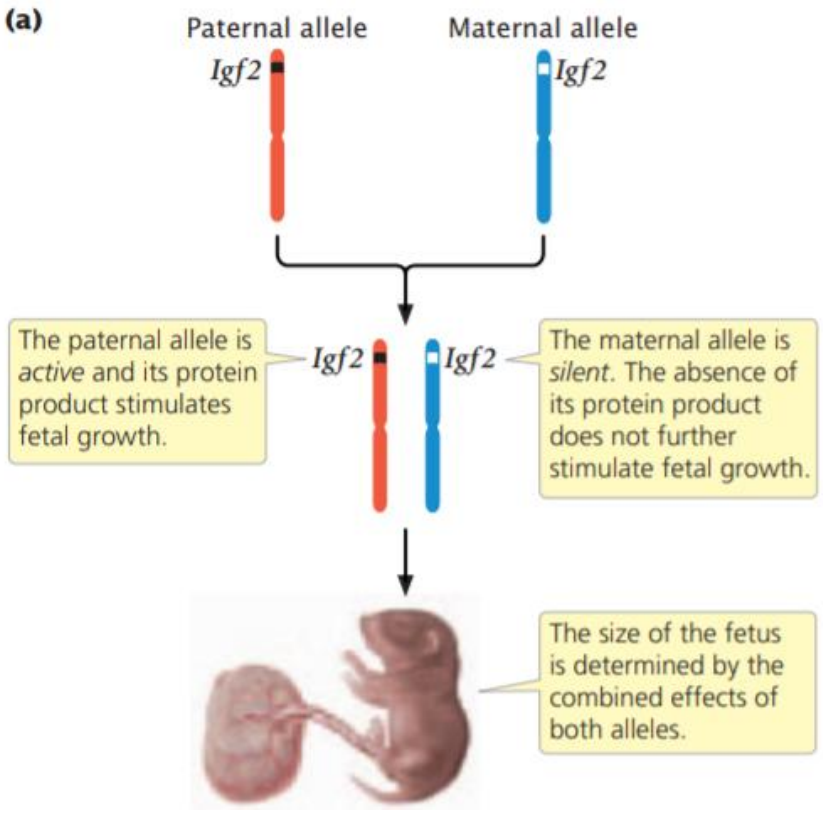
# Genomic imprinting and developmental control of methylation

- Experiments in the 1980s and 90s transplanting nuclei in mouse oocites proved that some of the cells' genes are imprinted with an indication of the gamete in which the genes began (Surani et al 1982)
- During gametogenesis, germ cells have their original methylation patterns reset based on the sex
- After fertilization, paternal and maternal genomes are demethylated and remethylated (except for differentially methylated regions associated with imprinted genes)
- In post-implantation stage, methylation becomes stage- and tissue-specific, with stable and lasting changes that define each individual cell type



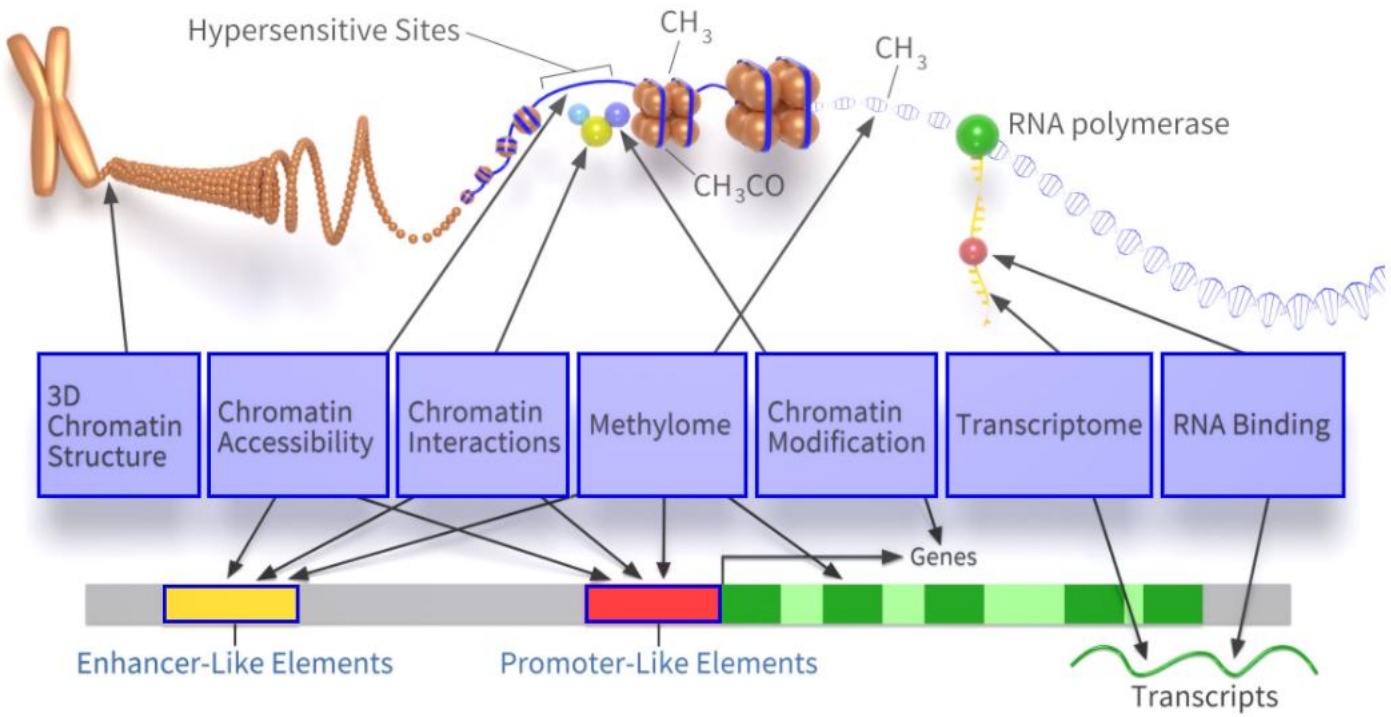
## The case of insulin growth factor 2 (*Igf2*)

- Offspring receives two alleles of the gene, but only the paternal one is expressed
- High expression of the gene increases size of the fetus
- Competing interests are fueling evolution of parental imprinting:
  - The paternal side wants to maximize the size of the offspring as it results in increased survival
  - The maternal side wants to balance the size of the offspring so to reduce energy investment



# The ENCODE project (Encyclopedia of DNA elements)

- launched by the US National Human Genome Research Institute (NHGRI) in September 2003 (after sequencing of the human genome)
- Genes amount to about 1.5% of the human genome. What about the rest?



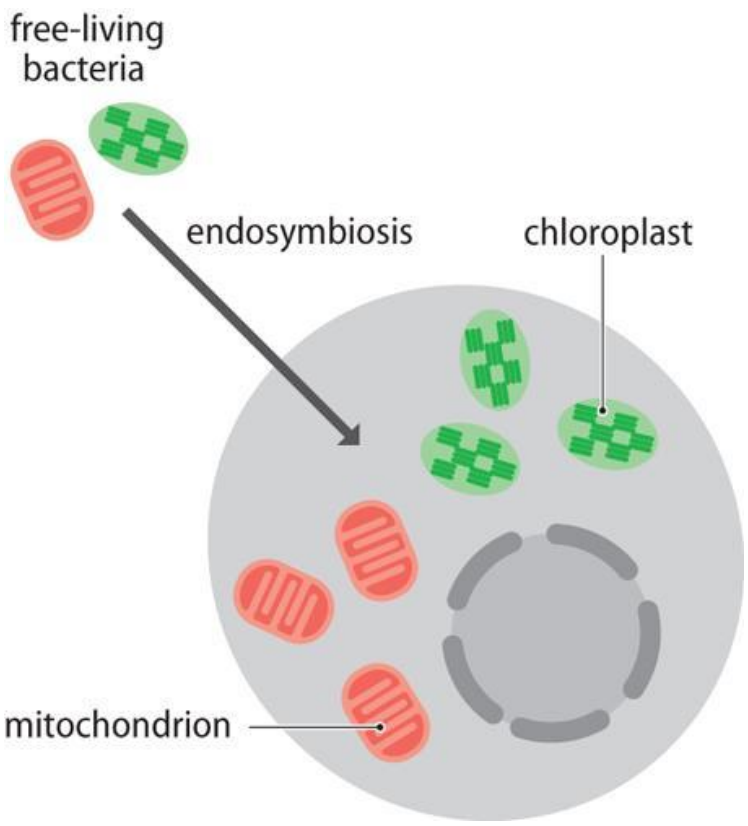

The ENCODE project discovered that 80% of the human genome has biochemical function

https://www.encodeproject.org/

# Other genomes in the eukaryotic cell

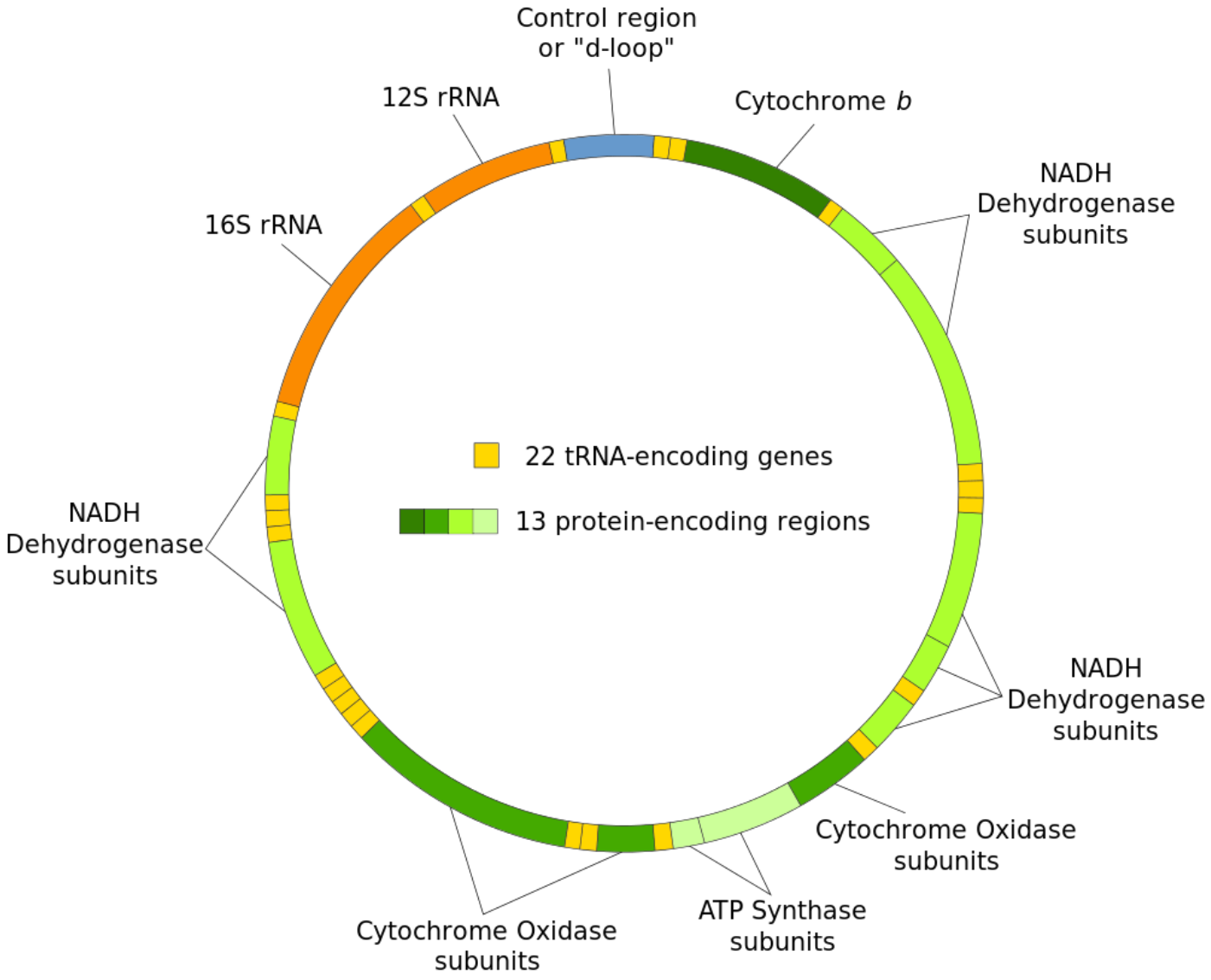
Mithocondria and chloroplasts may have originated as endosymbionts

- Gene expression processes occurring in organelles are similar in many respects to equivalent processes in bacteria
- Organelle genes are more similar to bacterias' than to those in the host cell



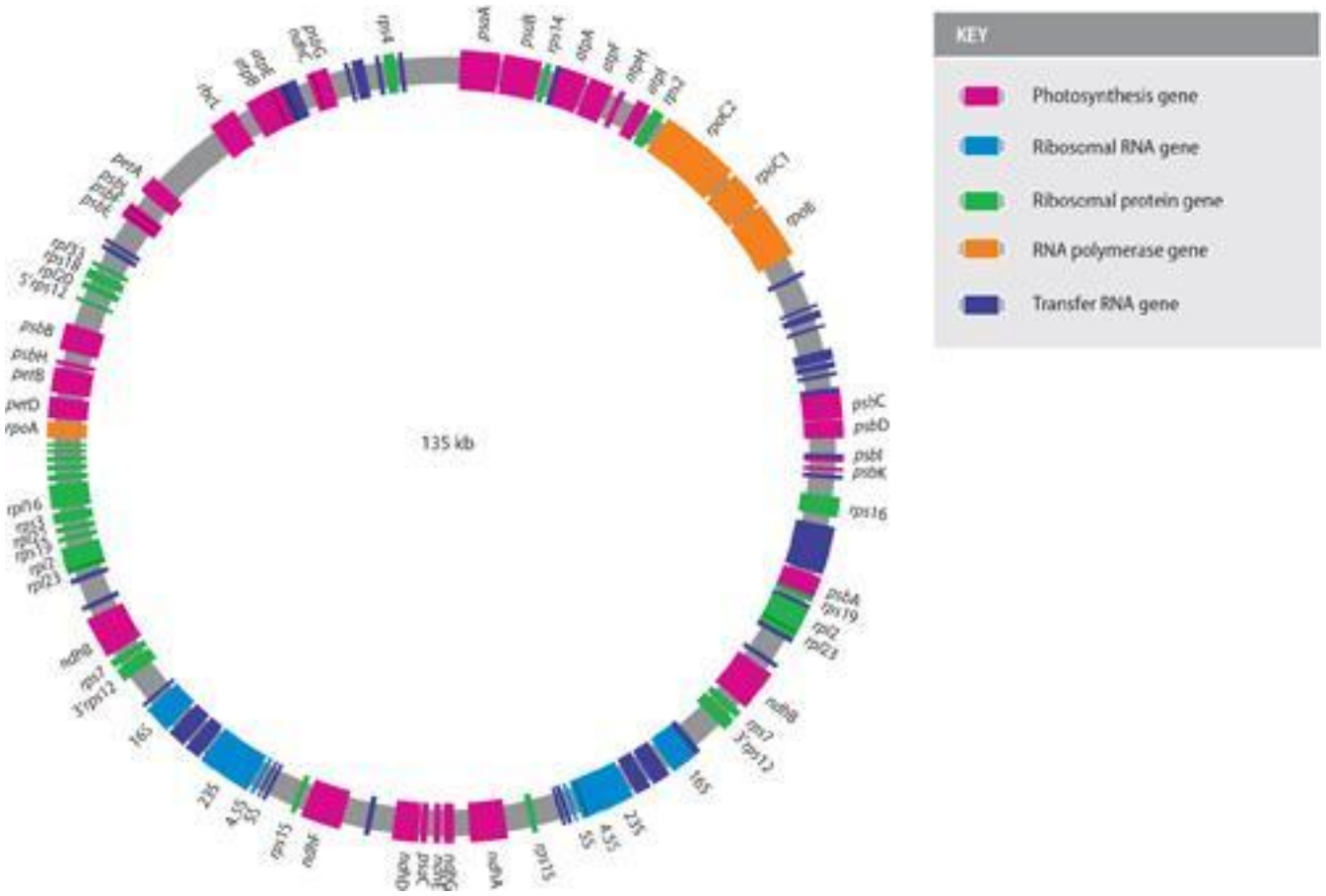
## **Mitochondrial DNA** (mtDNA)

- The living species most similar to mitochondria are members of two related orders: the Rickettsiales, which are obligate intracellular parasites, and the Pelagibacterales, which are free-living marine bacteria
- About 80 Kbp (*S. cerevisae*), encodes 13 protein ecoding regions
- Many other Mt proteins are actually produced in the cell nucleus, some are mantained (hydrophobic, colocalization for redox regulation)



**Chloroplast DNA** (cpDNA)

- The closest living relative is *Gloeomargarita lithophora*, a cyanobacterium inhabiting low-salt environments
- About 135 Kbp, approximately 120 genes (but changes depending on a large inverted repeat)
- All core components of the photosyntetic machinery + factors involved in their regulation
- About 10% of DNA in nuclei of land plants can be traced back to the chloroplast




Weird: in At, cpDNA is present in mtDNA, and both cpDNA and mtDNA are in the nucleus

- In any given diploid eukaryotic cell there's 2 copies of the nuclear DNA and 100-1000 copies of the organelles DNA.
- mtDNA size is not related to that of the host cell (bigger in plants than in animals)

| Species | Type of organism | Genome size (kb) |
|---|---|---|
| **Mitochondrial genomes** | | |
| *Plasmodium falciparum* | Protozoan (malaria parasite) | 6 |
| *Chlamydomonas leiostraca* | Green alga | 14 |
| *Mus musculus* | Vertebrate (mouse) | 16 |
| *Homo sapiens* | Vertebrate (human) | 17 |
| *Drosophila melanogaster* | Insect (fruit fly) | 19 |
| *Chondrus crispus* | Red alga | 26 |
| *Aspergillus nidulans* | Ascomycete fungus | 33 |
| *Reclinomonas americana* | Protozoa | 69 |
| *Saccharomyces cerevisiae* | Yeast | 86 |
| *Brassica oleracea* | Flowering plant (cabbage) | 360 |
| *Arabidopsis thaliana* | Flowering plant (vetch) | 367 |
| *Zea mays* | Flowering plant (maize) | 681 |
| *Cucumis sativus* | Flowering plant (cucumber) | 1556 |
| **Chloroplast genomes** | | |
| *Bigelowiella natans* | Chlorarachniophytes alga | 69 |
| *Marchantia polymorpha* | Liverwort | 121 |
| *Pisum sativum* | Flowering plant (pea) | 122 |
| *Oryza sativa* | Flowering plant (rice) | 135 |
| *Nicotiana tabacum* | Flowering plant (tobacco) | 156 |
| *Pelargonium spinosum* | Flowering plant | 217 |
| *Floydiella terrestris* | Green alga | 521 |
| *Haematococcus lacustris* | Green alga | 1350 |

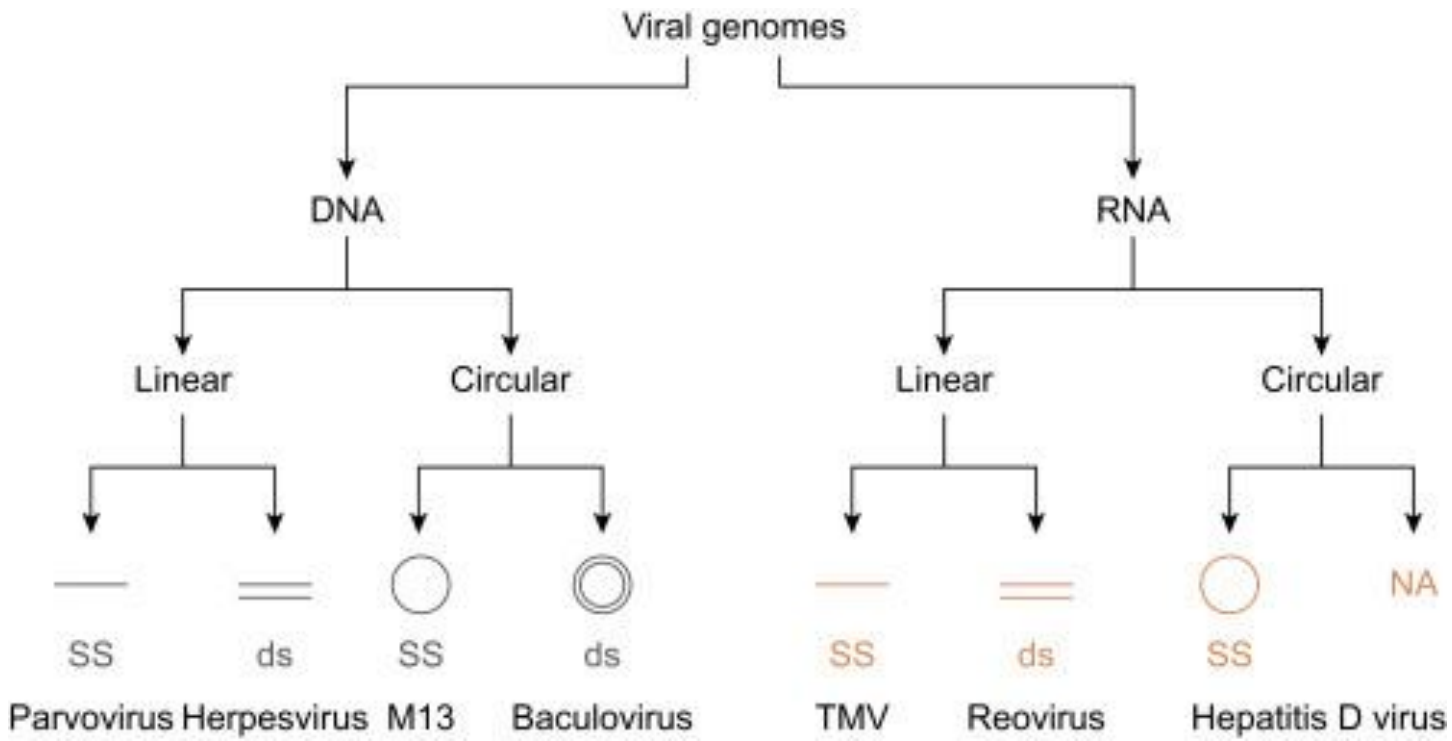
Data taken from NCBI Genome.

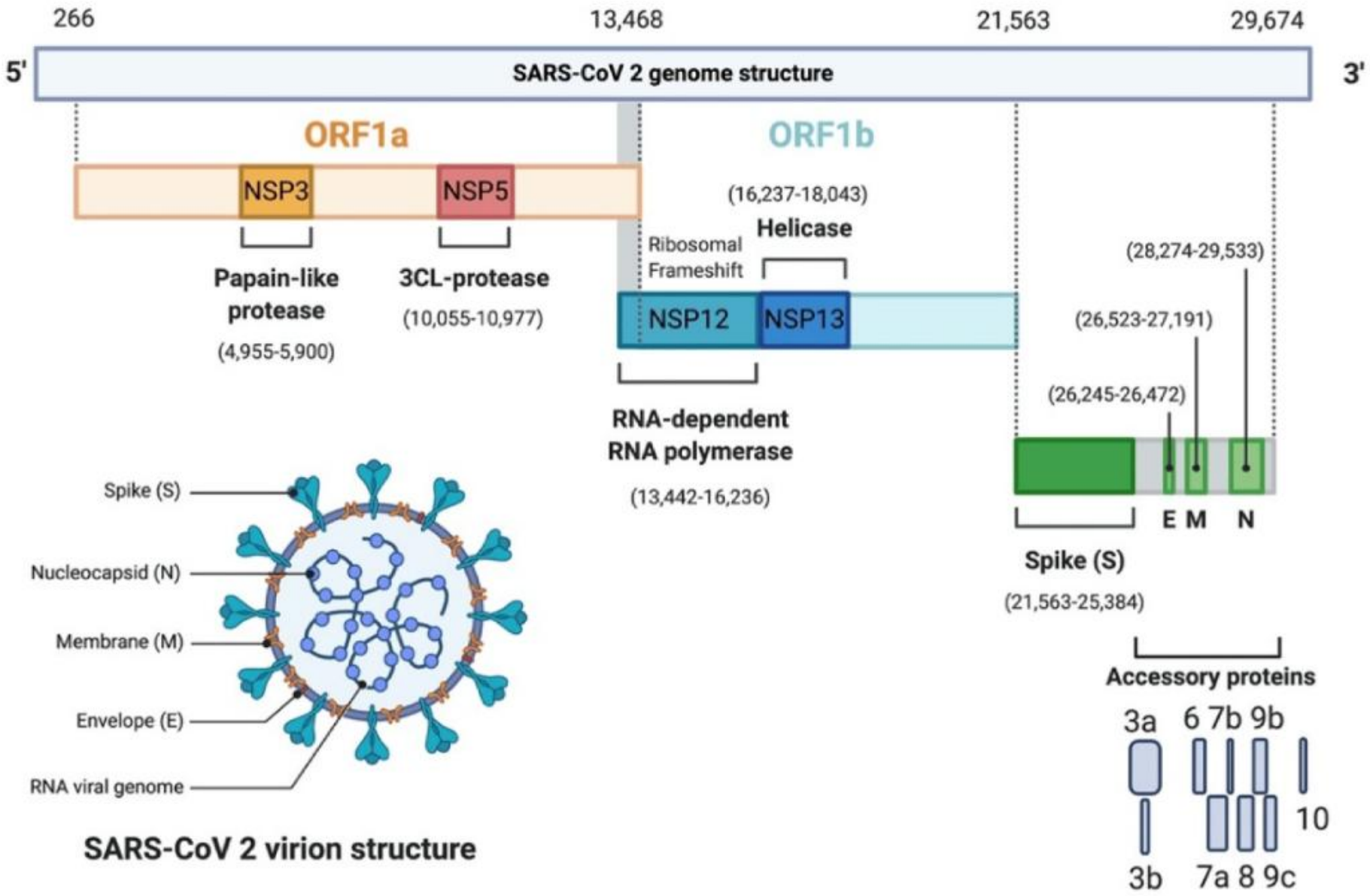
# Viral genomes

A bit of a weird beast; can be either DNA or RNA, circular or linear, double stranded or single stranded, positive-sense or negative-sense

Its whole business is doing two/three things:

- Coding proteins to aid in the creation of new virus particles
- (sometimes) Self-Replication
- Invading host cells, to reproduce and continue its life cycle

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NSP5
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Ribosomal Frameshift
(16,237-18,043)
Helicase
NSP12
NSP13
RNA-dependent RNA polymerase
(13,442-16,236)
(28,274-29,533)
(26,523-27,191)
(26,245-26,472)
E M N
Spike (S)
(21,563-25,384)
Accessory proteins
3a 6 7b 9b
10
3b 7a 8 9c
Spike (S)
Nucleocapsid (N)
Membrane (M)
Envelope (E)
RNA viral genome
SARS-CoV 2 virion structure

# Selfish DNA in the path of evolution

PNAS

## Multiple origins of viral capsid proteins from cellular ancestors

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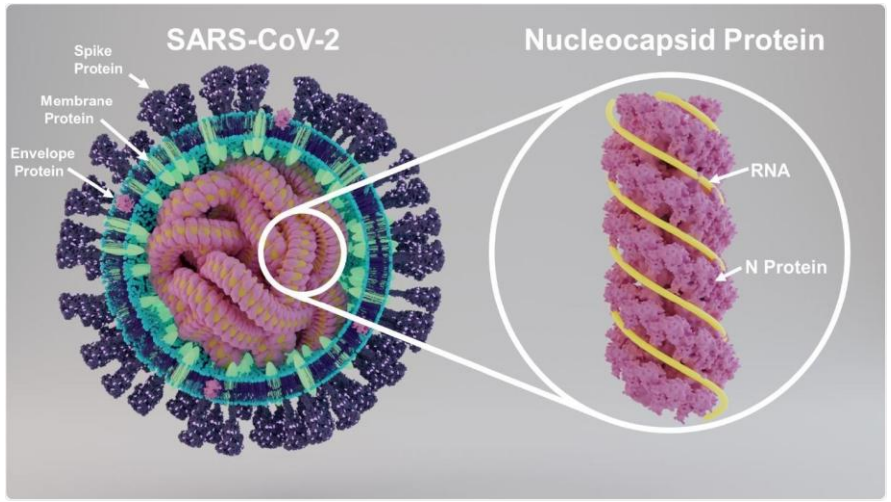
**Viruses are the most abundant biological entities on earth and show remarkable diversity of genome sequences, replication and expression strategies, and virion structures. Evolutionary genomics of viruses revealed many unexpected connections but the general scenario(s) for the evolution of the virosphere remains a matter of intense debate among proponents of the cellular regression, escaped genes, and primordial virus world hypotheses. A comprehensive sequence and structure analysis of major virion proteins indicates that they evolved on about 20 independent occasions, and in some of these cases likely ancestors are identifiable among the proteins of cellular organisms. Virus genomes typically consist of distinct structural and replication modules that recombine frequently and can have different evolutionary trajectories. The present analysis suggests that, although the replication modules of at least some classes of viruses might descend from primordial selfish genetic elements, bona fide viruses evolved on multiple, independent occasions throughout the course of evolution by the recruitment of diverse host proteins that became major virion components.**

genome replication. Understanding the origin o
possible only if the provenances of both compor
(11). Given that viral replication proteins oft
related homologs in known cellular organisms
suggested that many of these proteins evolved
world (4, 6) or in primordial, now extinct, cellu
13). The ability to transfer the genetic informat
capsids—the protective proteinaceous shells
cores of virus particles (virions)—is unique to b
distinguishes them from other types of selfis
such as plasmids and transposons (14). Thus, th
true viruses is inseparable from the emergence

Viral capsid proteins (CPs) typically do no
mologs among contemporary cellular protei
questions regarding their provenance and the c
which they have evolved. One possibility is that
have originated de novo within the genomes
replicons by generic mechanisms, such as ov

A widely accepted theory regarding the emergence and evolution of viruses highlights plasmids as self-replicating DNA elements that act in a self-serving manner.




Viral protein sequences seems not to be homologous to cellular proteins; and are different among virus types

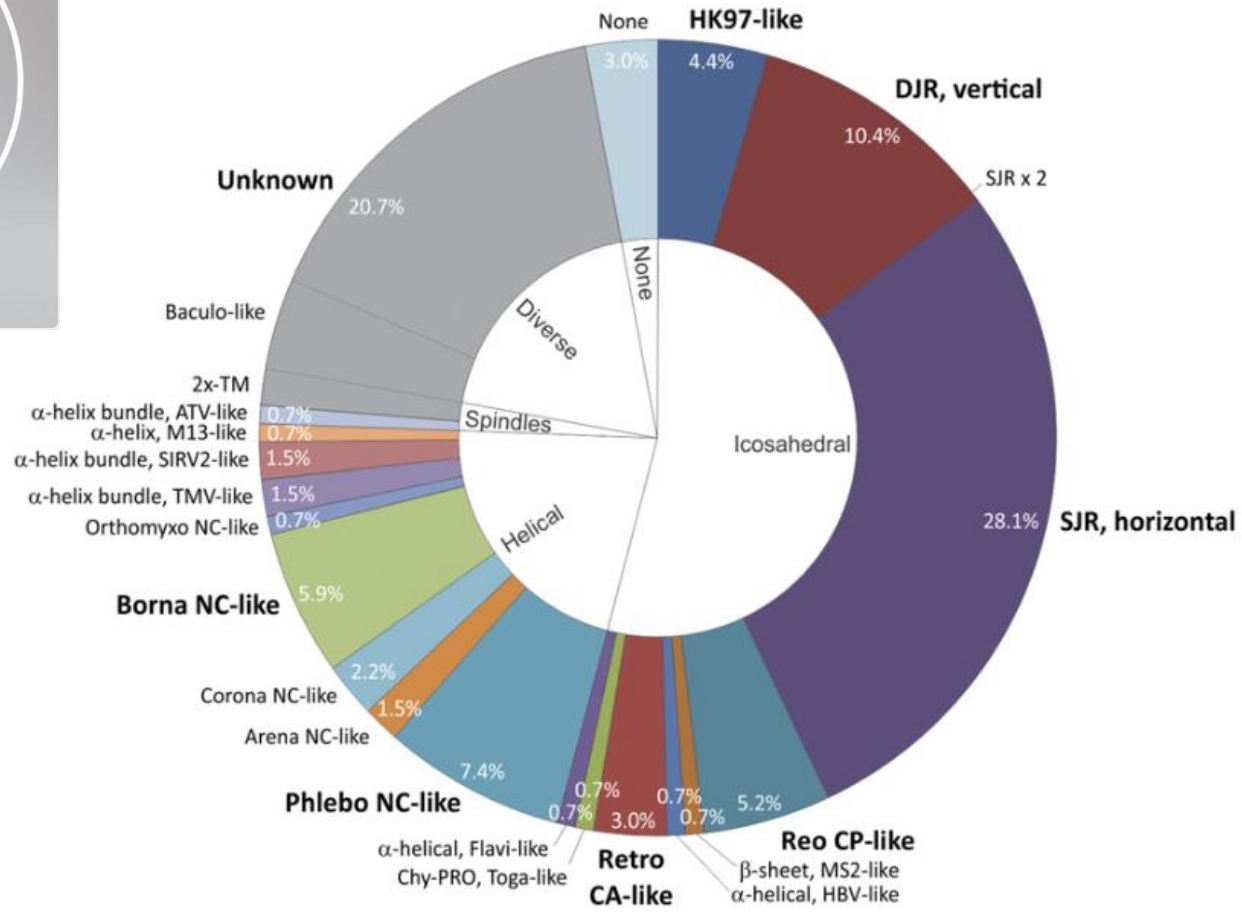
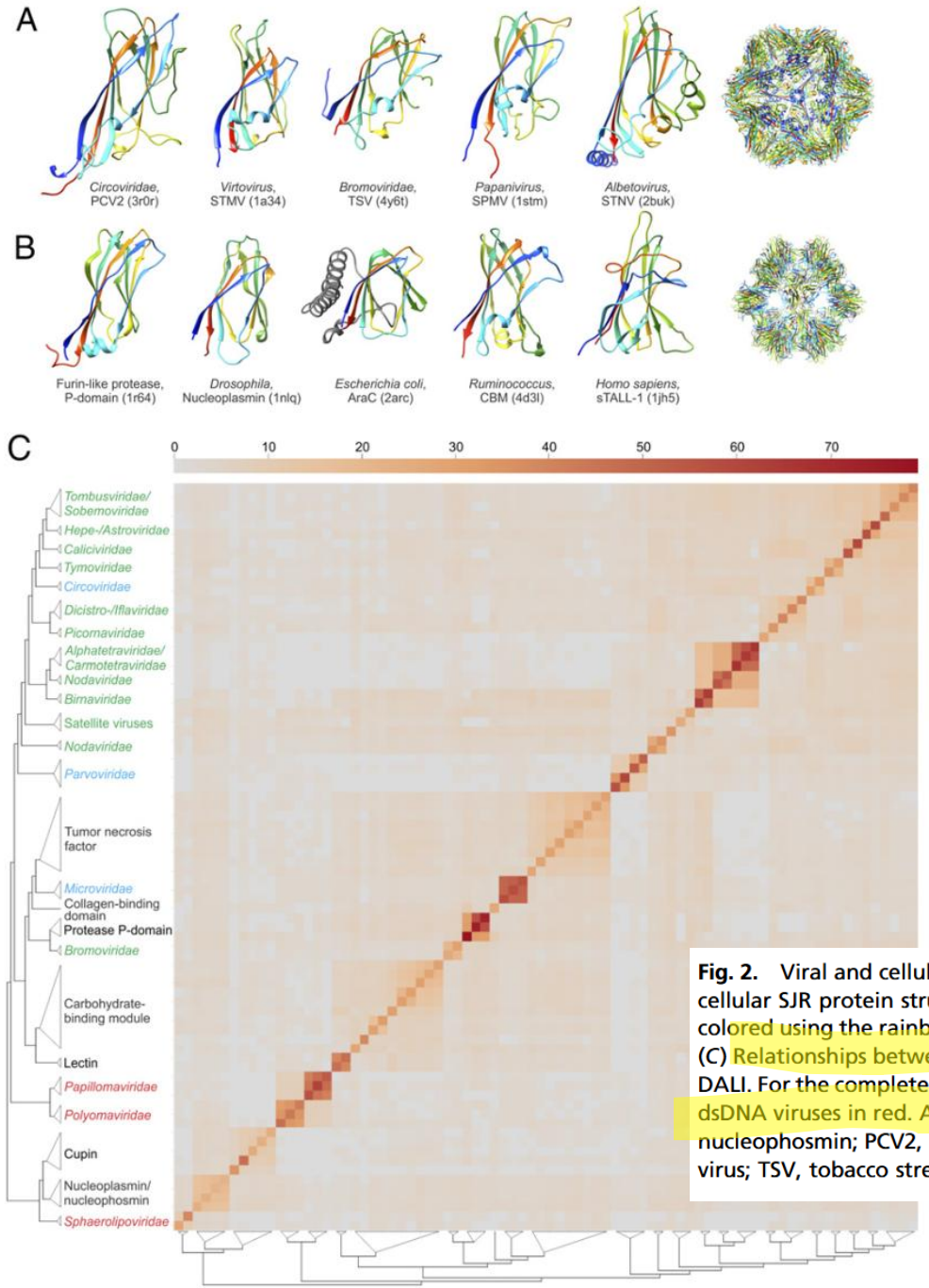
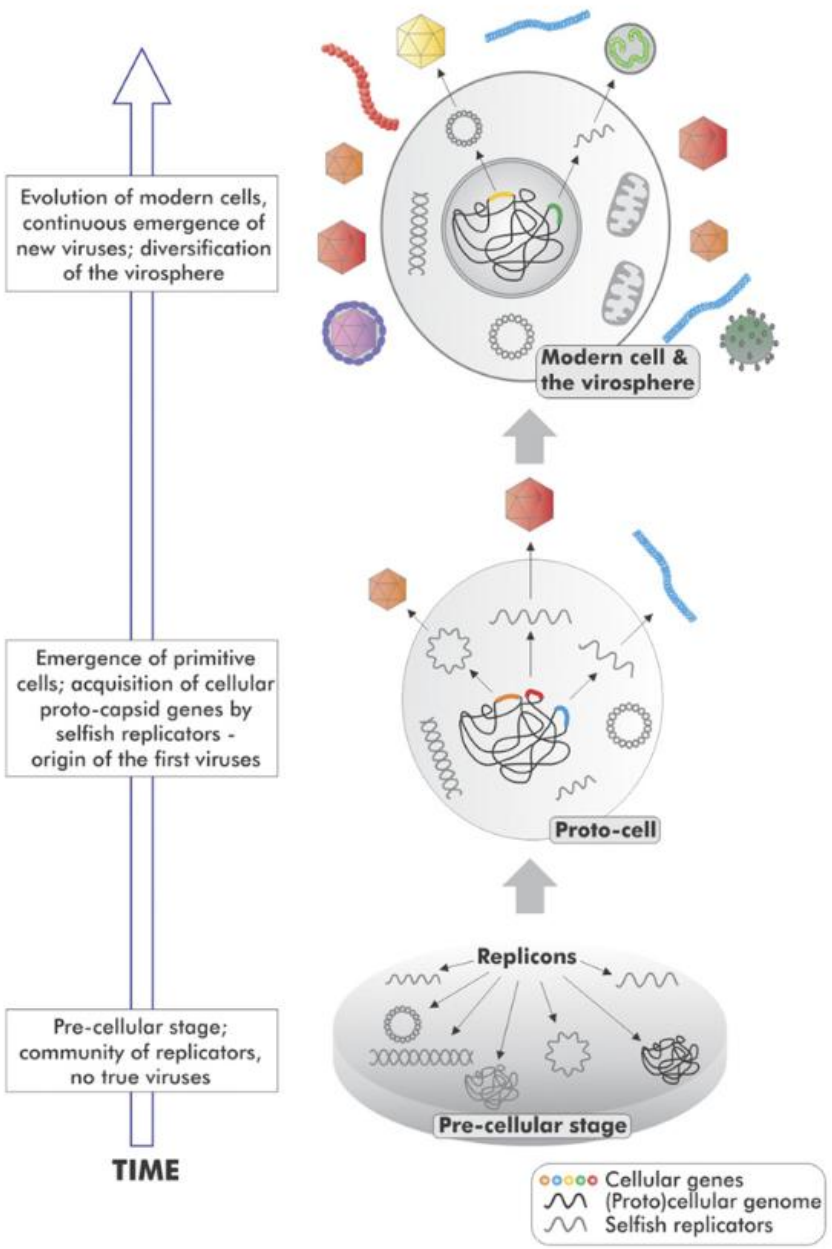



**Fig. 1.** Structural diversity of viral CP and NC proteins. The pie chart shows the distribution of architectural classes among 135 virus taxa (117 families and 18 unassigned genera; see Table S1). Arena, arenavirus; ATV, Acidianus two-tailed virus; Baculo-like, baculovirus-like viruses; Chy-PRO, chymotrypsin-like protease; Corona, coronavirus; CP, capsid protein; DJR, double jellyroll; Flavi, flavivirus; HBV, hepatitis B virus; NC, nucleocapsid protein; Orthomyxo, orthomyxovirus; Phlebo, phlebovirus; Reo, reovirus; Retro, retrovirus; SIRV2, *Sulfolobus islandicus* rod-shaped virus 2; SJR, single jellyroll; Toga, togavirus; TMV, tobacco mosaic virus; TM, transmembrane domain.



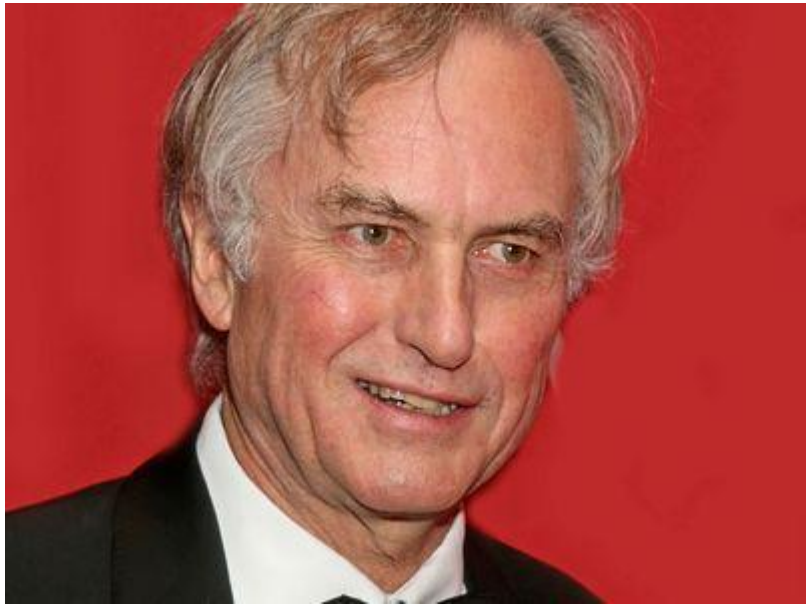

- Single Jellyroll (SJR) is the prevalent folding structure in all CPs (> 20%)
- It is possible to compare sequence (and predicted structure) in large databases of all living organisms

**Fig. 2.** Viral and cellular SJR proteins. (*A*) A selection of viral SJR CP structures. The rightmost structure corresponds to the virion of STNV. (*B*) A selection of cellular SJR protein structures. The rightmost structure corresponds to the 60-subunit virion-like assembly of the human sTALL-1 protein. All structures are colored using the rainbow scheme from blue (N terminus) to red (C terminus). The linker region leading to the DNA-binding domain in AraC is shown in gray. (*C*) Relationships between cellular and viral SJR proteins. The matrix and cluster dendrograms are based on the pairwise Z score comparisons calculated using DALI. For the complete matrix, see Dataset S1. The color scale indicates the corresponding Z scores. RNA viruses are shown in green, ssDNA viruses in blue, and dsDNA viruses in red. All compared structures are indicated with the corresponding PDB identifiers. CBM, carbohydrate-binding module; NP, nucleoplasmin/ nucleophosmin; PCV2, porcine circovirus 2; PRO-P, P domain of subtilisin-like proteases; SPMV, satellite panicum mosaic virus; STMV, satellite tobacco mosaic virus; TSV, tobacco streak virus.

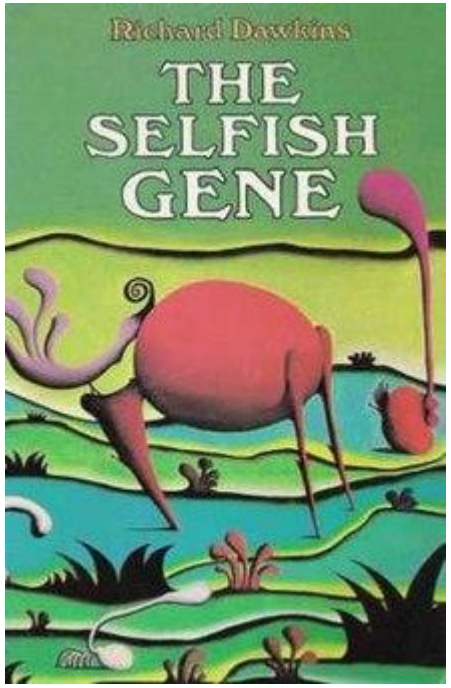



- The grand scenario for virus evolution becomes a hybrid between the virus-first and the escape hypotheses
- Favoring the pre-cellular hypothesis:
  - lack of close cellular homologs for the hallmark virus proteins
  - diversity of viral genomic strategies
  - early stages in the evolution of replicating genomes that implicate ensembles of small, partially autonomous, virus-like genetic elements
- Favoring the post-cellular hypothesis:
  - some of the primordial replicators coalesced and gave rise to the first cellular genomes
  - others became genetic parasites
  - viruses emerged through the recruitment of cellular carbohydrate-or nucleic acid-binding proteins as CPs
  - the simple, symmetrical, thermodynamically favored structures of capsids make it so that exaptation may not have been prohibitive.

# Are we selfish genes?

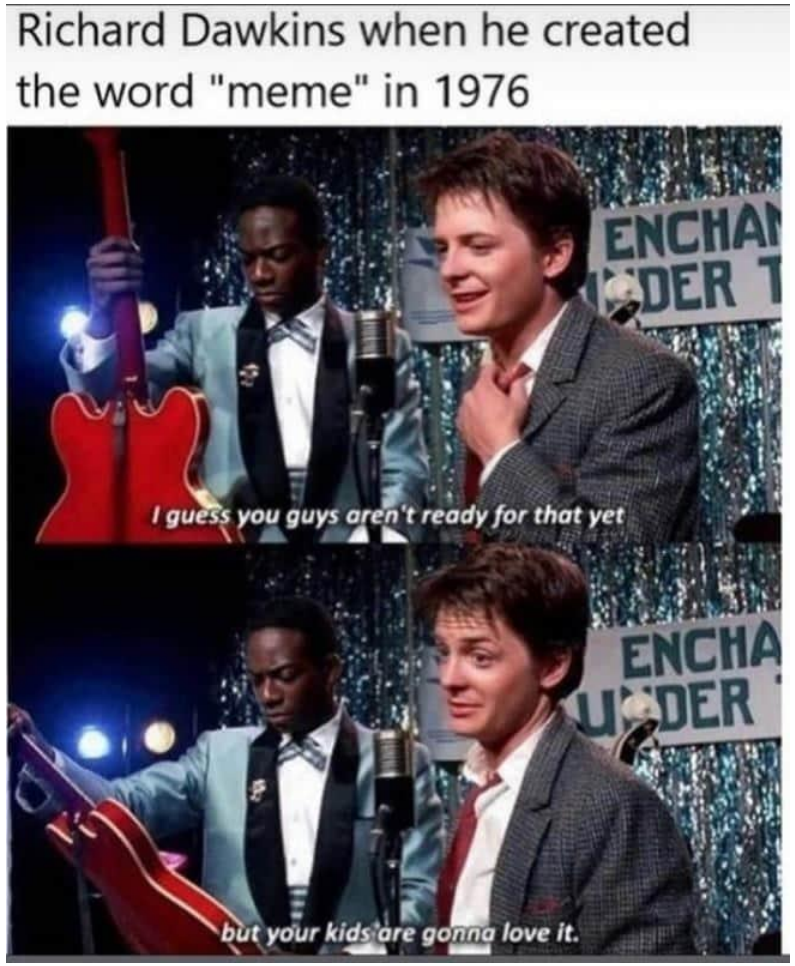


Richard Dawkins




A chicken is just an egg's way to make another egg

- 1976; Genes at the center of everything, living beings are vehicles
  - The genes are at the root of evolutionary processes
  - Organisms serve primarily as vessels for their genes, facilitating their replication
  - The better the gene controls the vehicle, the higher the chances to be passed on to the next generation
- Selfish Behavior
  - Organisms tend to preserve their own kind due to higher share of gene similarity

Meme, shortening of *mimeme* (from Ancient Greek "to imitate")

- meme as a “cultural gene”